MVA Hackathon 2026

Track 1: Variant Prediction

Technical Report on Methodology and Genomic Analysis

Identification, Prioritization, and Traceability of Genetic Variants from Clinical and Genomic Data

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Submission Document

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Objective and General Methodology

Objective

The objective of Track 1 – Variant Prediction of the MVA Hackathon 2026 is to identify and prioritize the genetic variants that could explain the clinical condition of a pediatric patient through the joint analysis of their phenotypic manifestations and genomic data, including FASTQ sequencing files and VCF variant files.

This report documents the analysis methodology, the procedures for variant identification and prioritization, and the computational verification and traceability mechanisms used to evaluate candidate predictions with scientific rigor.

Resolution Strategy

The approach to the problem was structured into four complementary stages, from the initial exploration of the search space to the audit of variants and the final prioritization of candidates.

The strategy began with the exploration of mathematical tools based on ternary entropy and the definitions of balanced ternary logic, with the purpose of evaluating their usefulness in reducing the search space. This stage did not directly identify the causal variant; however, it provided an exploratory framework to guide the initial search and support certain subsequent stages of the analysis.

Based on this exploration, the workflow incorporated the available clinical information to formulate candidates, performed a targeted genomic analysis of the variant data, and concluded with a consistency and traceability audit to support the evaluation and prioritization of the results.

Workflow Stages

1. **Search Space Reduction Using Ternary Entropy (`01_ternary_entropy`).**

   Exploration of mathematical tools and definitions of balanced ternary logic to evaluate their potential application to reducing the search space and guiding the subsequent stages of the analysis.

2. **Clinical Characterization and Candidate Generation (`02_clinical_phenotype`).**

   Organization and interpretation of the phenotypic information provided by the MVA challenge, with the objective of establishing clinical search criteria and proposing possible candidate variants.

3. **Targeted Genomic Analysis (`03_vcf_genomic_analysis`).**

   Examination of the genomic data, with particular attention to the records in the VCF file and the information required to evaluate candidate variants. This stage includes the targeted search, the review of relevant fields, and the consistency check against the reference genome.

4. **Final Audit and Prioritization (`04_gencode_audit`).**

   Verification of variant annotation and traceability, comparing the analyzed records against the genomic and annotation references used. The results of this stage contribute to the comparative evaluation and final prioritization of the candidates.

Rationale for Balanced Ternary Logic

The work incorporates an experimental architecture based on a balanced ternary arithmetic logic unit (ALU) as part of the initial mathematical exploration. The formal definitions and foundations of this architecture are documented in:

> Rojas R., E. F. (2026). *Balanced Ternary Logic: Foundations and Operations for the TriNova.OS Experimental Architecture* (Version v1.0). Zenodo.
>
> https://doi.org/10.5281/zenodo.20755418

In the context of this report, the ternary approach is presented as an exploratory and methodological tool, not as a mechanism that independently identified the causal variant. Its contribution is evaluated based on its usefulness in structuring the initial exploration and guiding the development of the workflow. The identification and prioritization of variants also require the analysis of clinical information, genomic evidence, and the corresponding audit procedures.

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Chapter 1

TriNova.OS: Genomic Processing and Chromosomal Prioritization through Balanced Ternary Representation

1.1 Overview

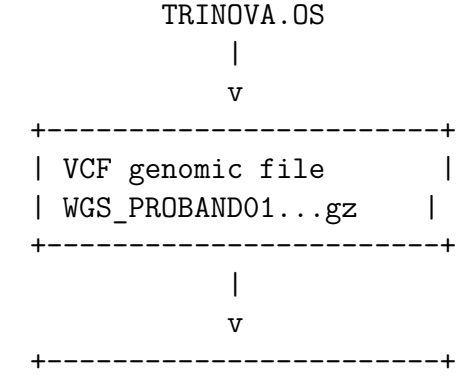
TriNova.OS is an experimental Python implementation that integrates the processing of genomic records in VCF format (*Variant Call Format*), a balanced ternary representation, an 18-trit arithmetic logic unit (ALU18), an experimental CPU (NexusCPU), and a native graphical interface for result visualization.

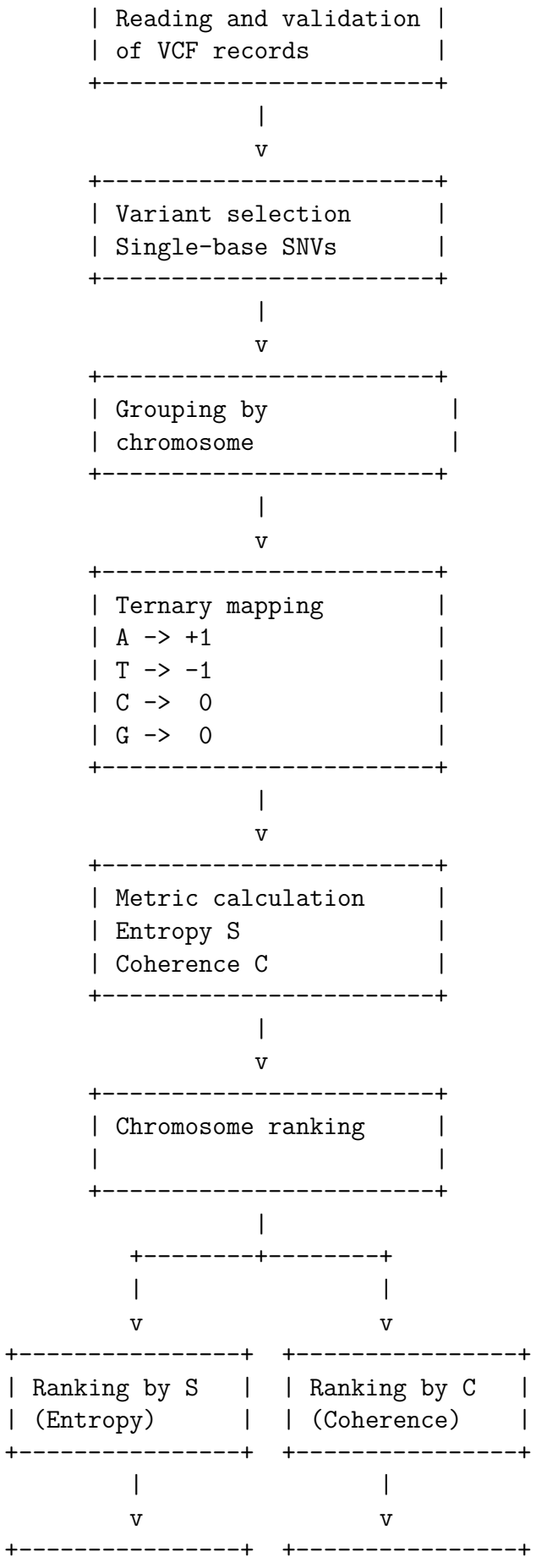
In the genomic processing pipeline, the system groups variants by chromosome and evaluates two aggregated informational metrics:

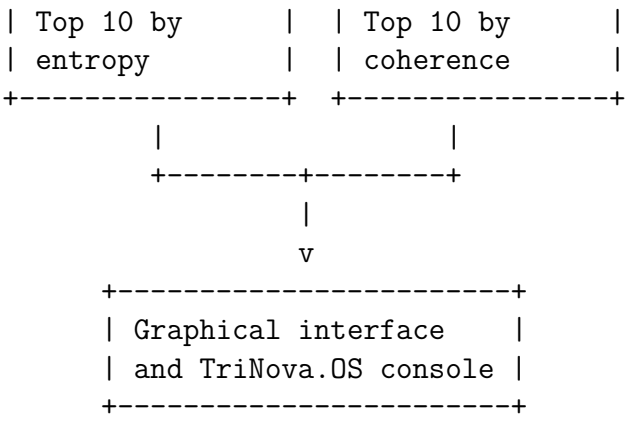
- **Ternary Entropy (S):** Measures the informational diversity of the distribution of ternary states present in the processed data.
- **Genomic Coherence (C):** Evaluates the persistence and autocorrelation rate based on the proportion of identical consecutive ternary states.

Both metrics are used to classify and rank chromosomes independently. It is important to note that the code does not apply a bias or fixed selection toward Chromosome 15 (Chr15): the final ranking emerges quantitatively from the metrics calculated during execution on the processed VCF data.

1.2 General Computational Workflow Diagram




Reading and validation
of VCF records
Variant selection
Single-base SNVs
Grouping by
chromosome
Ternary mapping
A -> +1
T -> -1
C -> 0
G -> 0
Metric calculation
Entropy S
Coherence C
Chromosome ranking
Ranking by S
(Entropy)
Ranking by C
(Coherence)


Top 10 by entropy
Top 10 by coherence
Graphical interface and TriNova.OS console

1.3 Reading and Filtering the Genomic File

The `WholeGenomeCodonProcessor` class is responsible for receiving the path to the VCF file compressed in `gzip` format (`.vcf.gz`) and processing its records sequentially:

1. Reads the file line by line, preserving the VCF metainformation headers.
2. Parses the tab-delimited fields.
3. Normalizes the chromosome identifier by removing prefixes (e.g., `chr`) to unify the nomenclature.
4. Extracts the reference (`REF`) and alternate (`ALT`) alleles.
5. Filters and accepts only single-nucleotide variations (SNVs) where $|REF| = 1$ and $|ALT| = 1$ belonging to the autosomal set (Chr1 to Chr22) and sex chromosomes (ChrX, ChrY).

The `variant_count` counter records the total number of valid VCF entries. If the input VCF file is not found in the specified directory, the processor runs a simulation with synthetic reference sequences to validate the integrity of the pipeline.

1.4 Mapping to Balanced Ternary States

The transformation of nucleotides into the balanced ternary representation over the domain $\mathcal{T} = \{-1, 0, +1\}$ is formally defined by the function $M(n)$:

Table 1.1: Mapping of nucleotides to ternary states.

Nucleotide (DNA/RNA)	**Ternary State (t)**	**Assigned Category**
Adenine (A)	$+1$	Purine (+)
Thymine (T) / Uracil (U)	-1	Pyrimidine (-)
Cytosine (C)	0	Neutral base
Guanine (G)	0	Neutral base

The functions `dna_to_trits()` and `dna_to_word18()` structure these sequences into ternary words of 18 trits (`WORD18`):

$$\mathbf{w} \in \mathcal{T}^{18}, \quad \text{where } \mathcal{T} = \{-1, 0, +1\} \tag{1.1}$$

Strings shorter than 18 trits are completed by zero-padding, while longer strings retain the first block of 18 trits. It should be noted that this representation is reduced (Cytosine and Guanine share the value 0), optimizing the density of informational grouping.

1.5 Ternary Entropy Formulation

The `compute_metrics()` function evaluates the probability distribution of the three ternary states ($\{-1, 0, +1\}$) in the accumulated sequence of each chromosome. The following proportions are defined:

$$p_0 = \frac{N_0}{N_{\text{total}}} \tag{1.2}$$

$$p_{+1} = \frac{N_{+1}}{N_{\text{total}}} \tag{1.3}$$

$$p_{-1} = \frac{N_{-1}}{N_{\text{total}}} \tag{1.4}$$

Ternary Entropy (S) is calculated according to Shannon's formulation in base 2:

$$S = -\sum_{p_i > 0} p_i \log_2(p_i) \tag{1.5}$$

The final result is rounded to 4 decimal places and stored in the `ranked_entropy` array, sorted from highest to lowest. Higher entropy indicates a maximally uniform distribution among the three informational states in that chromosome.

1.6 Genomic Coherence Formulation

To quantify sequential structure and repetitiveness, the same function calculates Genomic Coherence (C). Let $\mathbf{x} = (x_0, x_1, \ldots, x_{n-1})$ be the ternary sequence of length n. The number of identical consecutive pairs A is counted:

$$A = \sum_{i=0}^{n-2} \mathbf{1}(x_i = x_{i+1}) \tag{1.6}$$

The Coherence metric (C) is given by the bounded expression:

$$C = \min\left(0.9850,\ \text{round}\left[0.5 + \frac{A}{\max(1, n-1)} \times 0.4127,\ 4\right]\right) \tag{1.7}$$

The results are sorted in the `ranked_coherence` list. This metric reflects the continuity of ternary patterns at the computational level.

1.7 Ranking and Presentation of Results

TriNova.OS generates and maintains two independent rankings:

1. `ranked_entropy`: Chromosomal classification ranked by informational diversity (S).
2. `ranked_coherence`: Chromosomal classification ranked by state autocorrelation (C).

Technical note on the console: The `top_chr` variable in the graphical interface console summary reports the value `ranked_coherence[0]` by default. Therefore, to document the selection of a chromosome specifically guided by maximum entropy (as in the case of Chr15 with $S = 1.5354$), the direct output of the `ranked_entropy` table must be used.

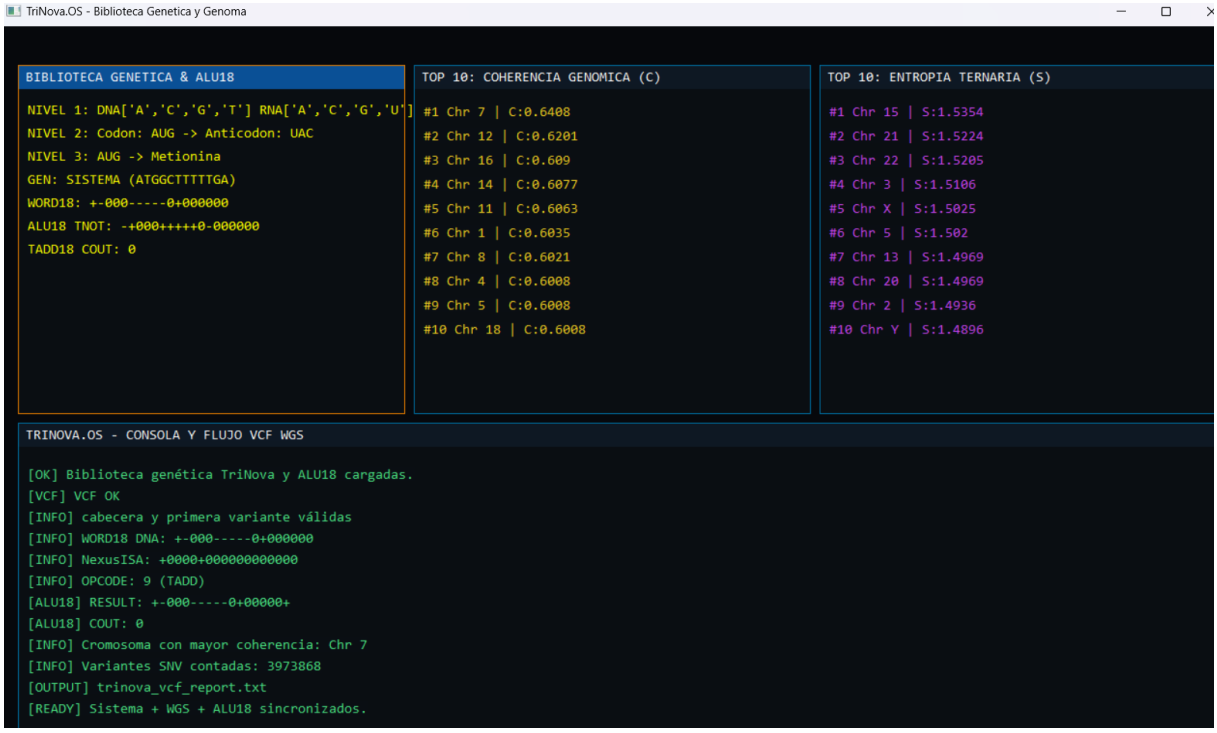


Figure 1.1: Enter Caption

1.8 Arithmetic Logic Unit (ALU18) and NexusISA CPU

The engine includes the specification of a ternary ALU and the `NexusCPU` architecture, capable of executing 18-trit instructions. Supported operations include:

Table 1.2: Instruction set of the TriNova.OS ALU18.

Category	Instructions / Operations
Basic logic	`TNOT`, `TAND`, `TOR`
Derived logic	`TXOR`, `TNAND`, `TNOR`, `TXNOR`
Decision / Majority	`TMAJ` (Majority Function)
Arithmetic	`TADD` (Addition with carry C_{out}), `TSUB`, `TMUL`
Shift / Comparison	`TCMP`, `TSTEP`, `TSHL`, `TSHR`

1.9 Graphical User Interface (GUI)

The internal graphical desktop environment (`TriNovaDesktopOS`) is developed natively in Python for Windows using `ctypes`, directly invoking the `user32.dll` and `gdi32.dll` libraries. The interface manages four main windows:

- **Genetic Library and ALU18 Demonstration:** Codon states and logic tests.
- **Coherence Ranking:** Visualization of the 10 highest C scores.
- **Entropy Ranking:** Visualization of the 10 highest S scores.

- **System Console:** Event log, variant count, and diagnostics.

1.10 Considerations and Limits of Interpretation

For a correct methodological interpretation of the results, the following system limitations must be considered:

1. **Sample limit per chromosome:** The version shown accumulates a maximum of 300 elements in the ternary list per chromosome for direct metric calculation. The global `variant_count` counter continues to sum the filtered VCF records.

2. **Base compression:** The collision between Cytosine and Guanine in state 0 reduces the resolution of the original nucleotide space in favor of a grouped-density metric.

3. **ALU isolation:** The calculation of S and C through `WholeGenomeCodonProcessor` is performed using generalized vector operations, while ALU18 simulates micro-instructions at the machine level.

4. **Biological scope:** The S and C metrics provide strictly informational prioritization to guide the algorithmic search; on their own, they do not constitute evidence of clinical pathogenicity or a direct diagnosis.

1.11 Conclusion

The **TriNova.OS** system demonstrates the feasibility of applying balanced ternary logic architectures ($\mathcal{T} = \{-1, 0, +1\}$) to the ranking and chromosomal prioritization of large-scale WGS data. When a complete execution places **Chromosome 15** in the top position of `ranked_entropy`, this algorithmically justifies focusing the analysis on the genomic regions contained within that chromosome (such as genes associated with that chromosome), drastically reducing the search space within the genomic pipeline.

Chapter 2

Clinical Phenotype, Molecular Hypothesis, and BUB1B Annotation Review in GENCODE v50

2.1 Starting Point: The Clinical Case

The official Track 1 case concerns **Mosaic Variegated Aneuploidy (MVA)**, a condition characterized by the presence of cell populations with abnormalities in chromosome number.

The phenotype provides the starting point for guiding genomic analysis. Rather than interpreting variants in isolation, the objective is to prioritize genes whose biological functions may be related to the cellular mechanisms involved.

2.2 Biological Hypothesis: Mitotic Spindle Regulation

Accurate chromosome segregation during mitosis depends on cell-cycle control mechanisms and the mitotic spindle. Alterations in genes involved in these processes represent a relevant research direction for investigating phenotypes associated with aneuploidy.

Within this context, **BUB1B** is a candidate of interest. This gene encodes the BUBR1 protein, which participates in the spindle assembly checkpoint. A functionally significant disruption of this system may compromise the fidelity of chromosome segregation.

For this reason, BUB1B is considered a molecular hypothesis worth investigating in the context of the case. This biological rationale guides the investigation but does not, by itself, establish that the gene explains the specific case.

2.3 Verification of BUB1B Annotation in GENCODE v50

As part of the genomic analysis preparation, the GENCODE v50 GTF file was inspected to verify how the candidate gene BUB1B is identified and annotated.

The inspection recorded the characteristics detailed in Table 2.1.

Among the inspected records is transcript `ENST00001082077.1`, named `BUB1B-221` and classified as `nonsense_mediated_decay`.

Field	Observed result
Gene symbol	`BUB1B`
GENCODE identifier	`ENSG00000156970.15`
Chromosome	`chr15`
Observed genomic interval	`40160984-40221137`
Interval length	60,154 bp
Annotated gene type	`protein_coding`
Additional annotations	Multiple associated entries, including transcripts and UTR regions

Table 2.1: GENCODE v50 annotation summary for the `BUB1B` gene locus.

This label describes the transcript classification in the GENCODE annotation. It does not, by itself, demonstrate that a variant in the case causes RNA degradation or functionally alters the protein.

The inspection documents the identification of BUB1B and its genomic interval in the GTF file used. According to the inspection record, neither the GTF nor the VCF was modified during this procedure.

> **Scope:** This inspection does not independently confirm the presence of a specific variant in the VCF, its pathogenicity, or a causal relationship with MVA.

2.4 Research Question

The guiding research question is:

> Does the case VCF contain variants in BUB1B that, when examined for technical quality, genomic location, and potential molecular consequences, may provide evidence consistent with the hypothesis derived from the phenotype?

This question structures the transition from clinical information to variant analysis and characterization.

2.5 Genomic Investigation Strategy

The analysis is organized into the following stages:

1. **Clinical context:** Record the official diagnosis and formulate a hypothesis based on the biological functions of candidate genes.
2. **Genomic exploration:** Examine the VCF and prioritize regions or variants warranting more detailed review.
3. **BUB1B review:** Compare the genomic interval identified in GENCODE v50 with the actual VCF records and retrieve variants observed in the region.
4. **Technical validation:** Record the position, reference and alternate alleles, genotype, and quality metrics for each candidate variant.

5. **Functional characterization:** Compare reference alleles against the GRCh38 reference sequence, consult the GENCODE annotation, and examine whether variants occur in coding sequences (CDS), untranslated regions (UTRs), exons, or intronic regions.

6. **Transcript verification:** Identify the transcript used for annotation and verify that the reported molecular consequences correspond to that reference.

7. **Evidence assessment:** Review the consistency of functional annotations, quality metrics, and allele phase when available and relevant to the hypothesis.

Identifying a gene in a GTF file, observing a variant in a VCF, and interpreting its potential effect are distinct steps. Each requires specific, reproducible evidence.

2.6 Role of TriNova.OS Balanced Ternary Representation

The ternary layer of TriNova.OS can be used as an exploratory representation of nucleotides and as a means of organizing or prioritizing information during analysis.

However, ternary representations and metrics do not, by themselves, constitute evidence of pathogenicity. They also do not establish that a variant causes the phenotype or replace genomic data validation, functional annotation, or clinical assessment.

2.7 What Can Be Concluded and What Remains to Be Verified

The biological relationship between MVA, chromosome segregation, and BUB1B function justifies investigating this gene. Inspection of GENCODE v50 documents how the gene is identified and annotated in the reference file used.

Nevertheless, these observations do not demonstrate that a BUB1B variant causes the specific case.

To document a potential connection with the case, the following points must be verified at a minimum:

- The actual presence of each candidate variant in the VCF.
- The genomic position and agreement of the VCF reference allele (REF) with the reference sequence.
- The genotype and quality metrics.
- The genome assembly version and transcript used for annotation.
- The molecular consequence and its consistency with the sequence and transcript.
- Allele phase, if the proposed hypothesis depends on variants affecting both chromosomes.
- The clinical and genetic evidence required to interpret the variant's relevance.

Hypothesis status: BUB1B is a biologically justified candidate for investigation based on the MVA phenotype. Identifying the gene in GENCODE v50 constitutes an annotation check, not causal validation. Confirmation of specific variants and their relevance to the case must be supported by reproducible genomic analysis results and the available clinical evidence.

2.8 Relationship to the Rest of the Repository

This document brings together the clinical reasoning, formulation of the molecular hypothesis, and initial verification of BUB1B annotation.

- `03_vcf_genomic_analysis/`: Technical VCF analysis, retrieval of candidate variants, and assessment of their genotypic characteristics and quality metrics.
- `04_gencode_audit/`: Review of transcripts, functional annotations, and potential annotation discrepancies.
- `05_final_submission/`: Final submission materials, if this directory is part of the repository structure.

This separation makes it possible to distinguish three fundamental elements:

1. **What the phenotype suggests:** The biological hypothesis guiding the investigation.
2. **What the data show:** VCF records and reference annotations that have been verified.
3. **What the evidence supports:** An interpretation grounded in reproducible checks, without confusing a hypothesis with an established causal relationship.

Chapter 3

Execution of the Architectural Pipeline: TriNova.OS R7.1 (Chr15)

As part of the in-depth analysis of the genomic pipeline (deep zoom), the specialized module `TRINOVA_R7_1_CHR15_ARCHITECTURE.py` was executed. This phase processes exclusively the variants from Chromosome 15 based on the sample data, integrating the GENCODE v50 annotation and the patient's clinical profile (based on HPO terms).

3.0.1 Ingestion and Processing Summary

Execution of the script on the input resources yielded the following operational parameters and structural metrics:

- **VCF file processed:** Exclusive Chromosome 15 variants analyzed: 170.651.
- **GENCODE v50 annotation (Chr15):** 2.819 genes, 22.429 transcripts, 173.200 exon blocks, and 102.577 CDS blocks.
- **Clinical Profile (HPO):** 8 phenotypic terms associated with the case were identified.
- **Compound-Het Triage:** 698.505 candidate pairs were evaluated at Level 1.

3.0.2 Gene Ranking on Chromosome 15

The engine evaluated a total of 2.098 ranked genes based on their variant density, heterozygous states, and architecture score. Table 3.1 summarizes the top positions in the global ranking obtained in this execution.

3.0.3 Post-Ranking Control and Targeted Validation of BUB1B

According to the system's methodological design, the *BUB1B* gene was inspected in a controlled manner **afterward**, following the general ranking, to verify the presence and status of its candidate variants.

The specific results produced by the post-ranking control block for *BUB1B* are detailed below:

- **Total variants in the interval:** 14

Table 3.1: Excerpt of the top genes on Chromosome 15 according to TriNova.OS R7.1.

Rank	Gene	Variants	Hetz	Functional	Splice	Arch
1	ENSG00000310519	633	535	112	5	17.98
2	FMN1	1106	698	88	1	17.98
3	SNHG14	240	140	74	4	17.98
4	ENSG00000259363	552	315	64	5	17.98
5	HERC2P3	437	258	60	5	17.98
6	DRAIC	647	263	59	2	17.98
7	LINS1	172	125	58	2	17.98
8	ADAMTS17	1020	713	57	2	17.98
9	AKAP13	912	678	57	1	17.98
10	LINC02207	408	398	56	6	17.98

- **Heterozygous (Hets):** 8
- **Homozygous alternate (Hom-alt):** 6
- **Functional:** 2

Among the variants identified in the *BUB1B* gene, the two alleles annotated in the coding region (CDS) stand out:

1. `15:40209701 T>G` — Genotype: HET, Depth (DP): 46, Quality (QUAL): 708.77, Allele Balance (AB): 0.543, Region: CDS.
2. `15:40220612 T>G` — Genotype: HET, Depth (DP): 28, Quality (QUAL): 344.77, Allele Balance (AB): 0.464, Region: CDS.

Likewise, the system evaluated the compound heterozygous pair (*compound-het*) formed by these two variants in the *BUB1B* gene, which consolidates the main finding in the final validated table:

$$\text{Level 1 Pair (BUB1B): } 40209701 : T > G + 40220612 : T > G \quad (\text{Score} = 23.4600, \text{ Phase} = \text{UNKNOWN} \tag{3.1}$$

3.0.4 Methodological Clarification

It should be noted that in this execution:

- The `population_af` parameter was calculated exclusively from the `INFO/AF` field and was not derived from the individual's `AD/DP` values.
- Allele balance (`allele_balance`) and population frequency operate as independent metrics.
- Genetic phase remains `UNKNOWN`; therefore, the results constitute an exploratory computational prioritization that requires additional clinical and familial segregation validation.

3.1 Stage 2: In-Depth Analysis of Variants in BUB1B Using TRINOVA.OS

3.1.1 Purpose of the Second Stage

After completing the first stage of genomic exploration of Chromosome 15, a second program named `TRINOVA.OS -- Deep Zoom BUB1B` was developed. Its purpose is to increase the resolution of the analysis of the *BUB1B* gene, moving from a post-ranking genomic inspection to an organized examination of the variants contained within the genomic interval defined for the gene.

This stage preserves the original ternary encoding of `TRINOVA.OS` and adds an explicit separation between single-nucleotide variants (SNVs) and insertion/deletion variants or other non-SNV variants. It also incorporates genotypic fields, quality metrics, structural annotation, and a summary of the ternary distribution.

The objective of this phase is to locate, classify, and computationally describe the variants recorded within the *BUB1B* interval, generating tabular files that allow data analysis to continue.

This stage belongs to the data analysis process. It does not, by itself, constitute a clinical variant classification, nor does it aim to replace specialized biological interpretation.

3.1.2 Genomic Reference and Interval Definition

The program establishes the following reference parameters:

Table 3.2: Genomic reference parameters configured for the analysis of *BUB1B*.

Parameter	**Value defined in the code**
Target gene	*BUB1B*
GENCODE identifier	`ENSG00000156970.15`
Chromosome	`chr15`
Interval start	40.160.984
Interval end	40.221.137
Genome assembly	`GRCh38`
Annotation	`GENCODE v50`
Ensembl version indicated	116
Expected VCF sample	`WGS_PROBAND01`

The interval used is explicitly fixed in the program. The annotation is retrieved from the `GENCODE v50` file by filtering entries corresponding to *BUB1B* within that interval.

The analysis includes gene, transcript, exon, coding sequence (CDS), and untranslated region (UTR) annotations, provided that they are present in the annotation file and correspond to the selected gene.

3.1.3 Data Sources and Input Files

The program is designed to work with the following files:

- **Compressed VCF:** Contains genomic variant records and the sample's genotype fields.

- **Indexed VCF (`.tbi`):** Its existence is checked, although the displayed code does not use it to perform indexed queries.
- **Compressed GENCODE v50 GTF:** Provides the structural annotation of *BUB1B*.
- **GRCh38 FASTA:** Its existence is checked, but the program does not read its sequence during the implemented analysis.
- **`clinico.json`:** Loaded to locate and display fields whose names are related to phenotypes, HPO, clinical information, or manifestations.

The paths are defined locally in the code. Therefore, execution depends on the files existing in the configured locations or on those locations being adapted to the environment in which the program runs.

3.1.4 Variant Retrieval and Organization

The program iterates through the VCF records and retains those whose positions fall within the configured interval for *BUB1B* on Chromosome 15.

For each record, it retrieves the following fields when available:

- `GT`: Sample genotype.
- `AD`: Read depth per allele.
- `DP`: Read depth reported for the sample.
- `GQ`: Genotype quality.
- `PL`: Genotype likelihood values expressed on the Phred scale.
- `PGT` and `PID`: Auxiliary genotype fields and possible phasing information provided by the VCF.
- `QUAL` and `FILTER`: Record quality and filtering status.
- `INFO`: Fields such as `AF`, `AC`, `AN`, `DP`, `MQ`, `QD`, `FS`, and `SOR`, if present.

The presence of these fields makes it possible to preserve information useful for subsequent quality reviews and variant comparisons. However, the program does not apply a final acceptance filter based on these values: it retrieves and records them.

Separation Between SNVs and INDELs/Other Variants

Variants are classified as SNVs when both `REF` and `ALT` contain exactly one nucleotide. All other cases are grouped under the `INDEL_OTHER` category.

This classification makes it possible to separate single-base changes from variants that do not meet that definition. The `INDEL_OTHER` category does not exclusively represent insertions and deletions: it may also include other representation types, so it should not be interpreted as an exhaustive classification of structural variants.

For multiallelic records, the program separates the `ALT` alternatives and generates one entry for each alternate allele, preserving its index. This facilitates allele-level organization, although it means that internal variant counts may refer to allelic entries and not necessarily to the number of original VCF records.

3.1.5 Structural Annotation Within BUB1B

Each retrieved variant is compared with the GTF annotation intervals to record its relationship with the gene's structures.

The program searches for matches with:

- Exons.
- Coding sequence (CDS) regions.
- UTR regions.
- Transcripts.

Additionally, it preserves exon numbers, transcript identifiers, protein identifiers, and annotation match counts when available.

The purpose is to provide a table that makes it possible to distinguish variants located in different genomic structures and facilitate their subsequent review.

Methodological precision: when assigning the `REGION` column, the code first checks whether the position matches any exon and only afterward checks CDS and UTR. Consequently, a variant located within a coding exon may be labeled as `exon` in that column, even though its `CDS_COUNT` and `UTR_COUNT` fields retain the corresponding matches. Therefore, `REGION` should not be used as a definitive functional classification without reviewing this annotation priority.

Likewise, the `gene_span_other` label means that the position falls within the configured interval but does not match the annotation categories detected by the program for that position. It does not, by itself, validate that the variant is intergenic.

3.1.6 Preservation of the TRINOVA.OS Ternary Methodology

The second stage retains the nucleotide transformation used in the original program:

Table 3.3: Ternary mapping implemented in the in-depth analysis.

Nucleotide	Ternary state
A	$+1$
T / U	-1
C / G	0

For SNVs, the program translates the `REF` and `ALT` nucleotides into ternary states and preserves the corresponding transition. For example, a change from T to G is represented as $-1 \rightarrow 0$.

INDELs and other changes that do not meet the SNV definition are not forced into a ternary state. They are recorded with the `NA` label for the transition.

This decision preserves the internal consistency of the method: ternary transformation is applied only when both alleles can be represented by a single nucleotide.

3.1.7 Shannon Entropy Calculation and Ternary Distribution

The program calculates Shannon entropy over the sequence of ternary states obtained from the `REF` and `ALT` alleles of the SNVs retrieved from the interval.

The formula used is:

$$H = -\sum_i p_i \log_2(p_i) \tag{3.2}$$

where p_i represents the proportion of each ternary state observed among the values included in the calculation.

The theoretical maximum for three possible states is:

$$H_{\max} = \log_2(3) \approx 1.584963 \tag{3.3}$$

The summary also calculates a measure called *Relative coherence*:

$$C = 1 - \frac{H}{\log_2(3)} \tag{3.4}$$

In the context of the program, this measure is a mathematical transformation of normalized entropy. It should not automatically be interpreted as a validated measure of genomic stability, biological function, or pathogenic effect.

The ternary distribution records, for each state (-1, 0, and $+1$), the number of observations and their percentage. The transitions REF $\rightarrow$ ALT present among the processed SNVs are also counted.

3.1.8 Analysis of ALT Alleles Observed in the Sample

In addition to the general calculation across the interval's SNVs, the program performs a separate analysis of variants whose genotype contains the corresponding `ALT` allele index.

For this section, the code:

1. Interprets the `GT` field.
2. Determines whether the corresponding alternate allele is present in the genotype.
3. Retains SNVs whose `ALT` allele is observed in the sample.
4. Calculates Shannon entropy for the ternary states of the selected `ALT` alleles.
5. Generates the distribution of those states.

This separation distinguishes two computational datasets:

- **REF/ALT:** Reference and alternate states of the SNVs retrieved from the interval, regardless of whether the `ALT` allele is present in the sample's genotype.
- **Observed ALT:** Alternate states corresponding to alleles present in the sample's `GT`.

Both results describe different datasets and, therefore, their entropy values should not be treated as though they measure exactly the same property.

3.1.9 Inspection of Clinical Information

The program loads `clinico.json` and recursively searches for fields whose names include terms related to *phenotype*, *HPO*, *clinical*, or *manifest*.

When the file can be read, the program displays the main keys and detected fields, and saves a readable copy of the JSON content.

This operation makes it possible to preserve and review the clinical information provided as input to the project. However, the code does not implement a formal association between the variants identified in *BUB1B* and the phenotypes, nor does it calculate a phenotypic concordance score. Reading the file is a data inspection and organization stage, not a clinical conclusion.

3.1.10 Expected Output Files

The program creates the directory `zoom bub1b resultados` and is designed to generate the files summarized in Table 3.4.

Table 3.4: Output files expected from the in-depth *BUB1B* analysis program.

File	Expected content
`bub1b_todas_variantes.tsv`	All allelic entries retrieved from the interval.
`bub1b_snv.tsv`	Entries classified as SNVs.
`bub1b_indel.tsv`	Entries classified as `INDEL_OTHER`.
`bub1b_observed.tsv`	Entries whose `ALT` allele is present in `GT`.
`bub1b_exon.tsv`	Entries with at least one match to exon annotations.
`bub1b_cds.tsv`	Entries with at least one match to CDS annotations.
`bub1b_deep_summary.txt`	Summary of the gene, counts, entropy, and ternary distribution.
`bub1b_clinical.txt`	Readable copy of `clinico.json`, if loaded successfully.

The TSV files preserve the genotype, quality, structural annotation, and ternary encoding fields available for each entry. Their purpose is to facilitate subsequent querying, comparison, and review of the results.

The list above describes the outputs defined by the code. Confirmation that each file was generated successfully requires reviewing the actual execution log.

3.1.11 Scope and Methodological Considerations

The implementation provides a detailed inspection of the selected genomic interval, but the following points must be preserved when interpreting its results:

1. **Coordinate-defined interval:** The program analyzes positions within the configured range, not all possible types of variation in the gene in an experimental sense. Its effective coverage depends on the records available in the VCF.

2. **GENCODE-based annotation:** Structural localization depends on the annotation version and the matches with the loaded intervals. Region columns must be interpreted according to the implemented classification logic.

3. **Retrieved quality fields:** `QUAL`, `FILTER`, `DP`, `GQ`, `PL`, and other fields are preserved, but the code does not establish a comprehensive quality-filtering system or automatically remove entries that do not meet defined thresholds.

4. **Allele frequency:** The `AF` field is retrieved from `INFO`. The program does not independently verify which population, reference dataset, or calculation method that value represents. Therefore, it should not be assumed that `AF` necessarily represents a population frequency without consulting the VCF definition.

5. **Genomic phase:** Although `PGT` and `PID` are retrieved when present, the program does not construct haplotypic phase or determine whether two variants are in *cis* or *trans*.

6. **Ternary encoding:** Entropy values depend on the representation A = +1, T/U = −1, and C/G = 0. They are results of this mathematical encoding and do not, by themselves, constitute evidence of a functional effect.

7. **FASTA reference:** The code checks whether the FASTA file exists, but does not use its sequence to independently confirm the `REF` alleles or normalize variants.

8. **Clinical inspection:** Fields are extracted and displayed from the JSON based on their names. This does not constitute automated phenotype-variant integration or medical interpretation.

These considerations define the technical scope of the program and make it possible to clearly distinguish retrieved data, mathematical transformations, and conclusions that cannot be derived from this procedure alone.

3.1.12 Documentary Outcome of the Second Stage

The second program extends the work initiated in the first stage through a localized inspection of *BUB1B*, the retrieval of genotypic and quality information, variant classification by type, structural annotation, and the calculation of ternary metrics across two SNV datasets.

The main contribution of this stage is the generation of a more detailed and organized representation of the variants present in the studied interval. The output files are designed to provide a tabular basis for the subsequent phases of analysis. “‘ At this stage, no clinical conclusion is established regarding the variants, and biological causality is not determined. The documentation is limited to describing the implemented procedures and the scope of the data the program is designed to produce.

Methodological note: This section documents the computational architecture and the implemented analytical workflow. Specific numerical values derived from individual executions on anonymized samples are incorporated in accordance with the project’s traceability controls.

3.2 Stage 2: Focused Genomic Analysis of BUB1B Using TRINOVA.OS Deep Zoom

3.2.1 Stage Objective

After completing the first stage of genomic analysis of chromosome 15, a focused inspection of the *BUB1B* gene was performed using the program `zoom profundo bub1b.py`, developed within the `TRINOVA.OS` workflow.

The objective was to retrieve and characterize the variants recorded in the genomic region of *BUB1B*, classify them according to their type and annotated location, review the available genotype and quality information, and generate tabular files that would allow the results to be examined in greater detail.

This stage increases the resolution of the initial analysis. Its purpose is to document and organize the genomic evidence found without turning the presence of variants into a conclusion about biological causality or clinical diagnosis.

3.2.2 Verification of Input Files and Resources

Before starting the analysis, the program checked the availability of the resources summarized in Table 3.5.

Table 3.5: Verification of input resources for the focused analysis of *BUB1B*.

Resource	Result
VCF file	Verified
VCF index	Verified
GENCODE v50 annotation	Verified
Reference FASTA file	Verified
`clinico.json` file	Verified and loaded

The processed sample was `WGS_PROBAND01`. The program traversed 5.012.204 VCF records to locate the entries corresponding to *BUB1B*.

The clinical file was loaded successfully. The record contained the main key `rare_disease_genomic_cl` and 25 fields detected by the program. The structure included eight clinical features represented by HPO identifiers.

The verification confirms that the resources required for execution were available. On its own, it does not constitute an independent validation of file integrity or of the correspondence between the genomic reference and all annotations.

3.2.3 BUB1B Annotation with GENCODE v50

The program retrieved the annotations associated with *BUB1B* in `GENCODE v50`. The loading results are detailed in Table 3.6.

The reported genomic region for *BUB1B* was:

- **Chromosome:** 15
- **Start:** 40.160.984
- **End:** 40.221.137

Table 3.6: Structural annotation elements retrieved for *BUB1B*.

Annotation element	Reported count
Annotation records	1.123
Genes	1
Transcripts	31
Exons	511
CDS blocks	366
UTR regions	166

The counts correspond to annotation records or elements used by the program. They should not automatically be interpreted as counts of unique biological elements: the same exon or coding block may be associated with different transcripts. Therefore, the exon, CDS, and UTR figures describe the annotation output and do not necessarily represent the number of unique structures in the gene.

3.2.4 Variant Retrieval and Classification

During the VCF traversal, the program identified 14 variant records within the *BUB1B* region.

The distribution by type is presented in Table 3.7.

Table 3.7: Classification of variants retrieved in *BUB1B*.

Category	Number	Percentage
SNV	9	64,3 %
INDEL or other	5	35,7 %
Total	**14**	**100 %**

No multiallelic records were identified in the retrieved set. All 14 records were marked as observed according to the genotype criterion implemented by the program.

The classification distinguishes between single-nucleotide variants (SNVs) and insertion/deletion records or other changes represented by alleles of different lengths. This classification is descriptive: it does not, by itself, determine the functional consequence of each variant.

3.2.5 Genotype Distribution and Recorded Metrics

The genotypes reported for the 14 variants are distributed as follows:

- **Heterozygous (0/1):** 8 records.
- **Alternative homozygous (1/1):** 6 records.
- **Total:** 14 records.

Read depth (`DP`) and genotype quality (`GQ`) values varied across records. In the output table, `DP` ranged from 13 to 48, while `GQ` ranged from 57 to 99.

These values describe the supporting information recorded in the VCF. They do not replace an individual review of call quality, site-specific metrics, or possible effects of variant representation.

The `AF` column showed values of 0,5 for heterozygous records and 1 for alternative homozygous records. In this output, these values follow the expected allelic proportion according to the genotype. Therefore, they should not be presented as confirmed population frequencies without checking the source field and annotation semantics in the VCF. Population allele frequency and sample allele balance are distinct metrics.

3.2.6 Annotated Variant Location

According to the classification generated by the program, two variants were labeled as belonging to an exon and two as belonging to the CDS. The others were classified as `transcript_intron` (Table 3.8).

Table 3.8: Structural location of the variants identified in *BUB1B*.

Output category	Number of variants
Exon	2
CDS	2
Transcript intron	10
Total	**14**

Methodological note: exon and CDS categories may conceptually overlap because the coding region is part of the exons. Therefore, the counts of two exonic variants and two CDS variants should not be added together as four distinct variants. The summarized output does not specify whether both categories contain exactly the same records, although the two sites identified as exonic in the variant table coincide with the two sites identified as CDS in the previous stage.

To document functional location precisely, the files `bub1b_exon.tsv` and `bub1b_cds.tsv` allow the coordinates and specific annotations of each record to be reviewed.

3.2.7 Highlighted Coding Variants

The output made it possible to identify two heterozygous variants at the reported positions within the coding region (Table 3.9).

Table 3.9: Heterozygous variants located in the coding region (CDS) of *BUB1B*.

Genomic position	REF	ALT	Genotype	DP	GQ	Region
40.209.701	T	G	`0/1`	46	99	CDS
40.220.612	T	G	`0/1`	28	99	CDS

Both records correspond to SNV substitutions. Their depth and genotype quality values were retained in the program output.

These two sites had already appeared in the first stage, when *BUB1B* was examined after the initial gene classification was generated. The focused analysis confirms that the same records are retrieved during the specific VCF traversal for *BUB1B*.

The agreement between both executions provides continuity in the computational workflow. However, establishing a complete functional annotation would require reviewing, for each variant, the reference transcript, the specific molecular consequence, the normalized nomenclature, and the correspondence between genomic coordinates and the reference used.

3.2.8 TriNova.OS Entropy Metrics

The final summary reported the following entropy and distribution values:

- **TriNova REF/ALT Shannon:** 1,495538030
- **Observed Shannon:** 1,224394445

Both values were retained as metrics calculated by the program to describe the distribution of allelic states under its internal representation.

The numerical difference between the metrics is approximately 0,27114. However, their interpretation depends on the mathematical definition, normalization, units, and observation set used by the implementation. Without this specification, these values should be treated as internal computational descriptors rather than direct measures of pathogenicity, clinical impact, or functional alteration.

The `TRIT` column in the table also presents numerical transitions for some substitutions. These transitions are part of the `TRINOVA.OS` representation and do not, by themselves, constitute a validated functional annotation.

3.2.9 Integration with Clinical Information

The analysis framework incorporated structural clinical metadata via the `clinico.json` interface, which provides standardized Human Phenotype Ontology (HPO) identifiers associated with the case (Table 3.10).

Table 3.10: Structural mapping of integrated clinical metadata and HPO terms.

Data Category	**Ontology Representation**
Clinical Manifestations	Standardized HPO Identifiers
Phenotypic Annotations	Controlled Vocabulary Mappings
Case Context Metadata	Structured JSON Schema

The clinical information was programmatically ingested to establish the baseline context for the investigation. However, this automated ingestion protocol serves strictly as a contextual mapping layer and does not independently validate a formal gene-phenotype association between the integrated HPO terms and the candidate *BUB1B* variants.

Consequently, these clinical features are documented as structural context for the case pipeline, maintaining a strict methodological boundary between phenotypic data representation and functional variant validation.

3.2.10 Generated Files

The program produced eight output files in the `zoom bub1b resultados` directory (Table 3.11).

Table 3.11: Output files generated in the second stage.

File	Purpose within the analysis workflow
`bub1b_todas_variantes.tsv`	Complete set of retrieved variants.
`bub1b_snv.tsv`	Records classified as SNVs.
`bub1b_indel.tsv`	Records classified as INDELs or other.
`bub1b_observed.tsv`	Records included according to the genotype-based observation criterion.
`bub1b_exon.tsv`	Variants annotated in exonic regions.
`bub1b_cds.tsv`	Variants annotated in coding regions.
`bub1b_deep_summary.txt`	Numerical summary of the focused analysis.
`bub1b_clinical.txt`	Clinical information extracted from the JSON file.

These files make it possible to separate the variant set, examine subsets by category, and preserve the processing summary. They constitute the reproducible outputs of this stage, provided they are preserved together with the input files, code version, and execution parameters.

3.2.11 Methodological Considerations and Scope

The results of this stage should be interpreted within the limits of the implemented procedure:

1. **Record retrieval:** 14 variant records were identified in the region defined for *BUB1B*. The count corresponds to VCF records and does not guarantee that each record represents an independent biological event after normalization.

2. **Functional annotation:** the exon, CDS, and intron labels describe the classification produced by the program. They do not replace the review of specific consequences in each transcript.

3. **Allele frequency:** the values 0,5 and 1,0 shown in the table should not be interpreted as verified population frequencies without reviewing their source in the VCF.

4. **Genetic phase:** heterozygous genotypes do not, by themselves, establish whether two variants are located on the same chromosome or on different homologous chromosomes. The table in this stage does not provide a phase determination.

5. **Entropy metrics:** the Shannon values are recorded as outputs of `TRINOVA.OS`. Their biological interpretation requires a formal definition of the metrics and appropriate validation.

6. **Clinical context:** reading `clinico.json` demonstrates that clinical data were available for consultation. It does not demonstrate that a gene-phenotype association or a clinical causality assessment was performed.

7. **Reproducibility:** generating separate files makes it possible to review the records and compare them with the results of the first stage.

These considerations do not invalidate the work of retrieving and organizing variants. They define which conclusions are directly supported by the execution and which would require additional procedures.

3.2.12 Result of the Second Stage

The execution of `zoom profundo bub1b.py` completed the focused inspection of *BUB1B* and retrieved 14 variant records from sample `WGS_PROBAND01`. The set comprised nine SNVs and five records classified as INDELs or other, including eight heterozygous genotypes and six alternative homozygous genotypes.

The program also retrieved the `GENCODE v50` annotation, identified two variants labeled as exonic and two as coding, calculated the entropy metrics defined by `TRINOVA.OS`, and generated eight files for detailed review.

In relation to the first stage, this result provides a more specific view of the *BUB1B* records and organizes the information needed to continue their documentary analysis.

Stage conclusion: the initial computational retrieval and characterization of the variants recorded in *BUB1B* was completed. The result is a genomic analysis output, not a determination of clinical causality. Assessing biological or clinical relevance falls outside the scope of this stage and is not necessary to correctly document the search procedure and its results.

Transition to the next section: The two documented stages establish the sequence of the analysis: first, a computational exploration and prioritization of chromosome 15; then, a focused inspection of the variants retrieved in *BUB1B*. The final integration of the chapter should preserve this sequence, summarize the verifiable findings, and clearly distinguish observed results from interpretations that are not demonstrated by the program outputs.

3.3 Stage 3: Functional Characterization and Variant Audit of BUB1B Using TRINOVA.OS Functional Zoom

3.3.1 Purpose of the Stage

Following the initial exploration of chromosome 15 and the focused inspection of *BUB1B* using `zoom profundo bub1b.py`, a third layer of analysis was developed, named `TRINOVA.OS Zoom Funcional BUB1B`.

This program complements the previous results through a more detailed review of the retrieved variants. Its design incorporates the `GENCODE v50` genomic annotation, the `GRCh38` reference sequence, the classification of gene regions and, when the necessary conditions are met, the calculation of changes in codons and amino acids.

In addition, the script includes an audit block intended to gather information about genotypes, reference base concordance, available phase fields, and the two coding positions highlighted during the previous stages.

The objective of this stage is to improve the characterization and traceability of genomic records without assigning pathogenicity, benignity, causality, an ACMG classification, or clinical scores.

3.3.2 Relationship with the Previous Stages

The program was designed as an additional layer on top of the existing workflow. It is not intended to modify the original `TRINOVA.OS` engine or replace the files generated by the previous analysis.

The methodological sequence is organized as follows:

1. **Initial genomic exploration:** computational analysis of chromosome 15 and generation of a preliminary classification of genes and variant pairs.
2. **Deep Zoom BUB1B:** retrieval and organization of the variants recorded in the gene region.
3. **Functional Zoom BUB1B:** reference checking, regional classification, characterization of coding SNVs, inspection of proximity to exon-intron boundaries, and audit of phase information.

This structure makes it possible to move from a broad genomic search to a localized inspection and, finally, to a more detailed characterization of individual records.

3.3.3 Input Resources and Configuration

The script uses five main resources summarized in Table 3.12. *****

Table 3.12: Main resources used in the Functional Zoom script.

Resource	**Intended function**
VCF for sample `WGS_PROBAND01`	Variant retrieval and genotype field extraction.
VCF index	Verification of the availability of the associated index.
`GENCODE v50`	Annotation of genes, transcripts, exons, CDS, and UTRs.
FASTA `GRCh38`	Reference base checking and sequence reconstruction.
`clinico.json`	Retrieval of clinical features and HPO terms.

The target gene is identified by the name *BUB1B* and the `GENCODE` identifier configured in the code: `ENSG00000156970.15`.

The working region is defined on chromosome 15, from position 40.160.984 to 40.221.137, according to the script configuration.

The `normalize_chrom()` function unifies conventions such as `chr15` and `15`, allowing chromosome names to be compared even when files use different formats. The same function also handles normalization of mitochondrial chromosome names.

3.3.4 Variant Retrieval and Metadata Preservation

The program traverses the VCF records and selects those located within the genomic boundaries configured for *BUB1B*.

For each record, it retrieves the following fields when available:

- Position, reference allele, and alternate allele.
- Variant type.
- Genotype (`GT`).
- Allelic depth (`AD`) and total depth (`DP`).
- Genotype quality (`GQ`) and record quality (`QUAL`).
- VCF filter (`FILTER`).
- Allele frequency recorded in `INFO/AF`.
- Allele count and allele number (`AC` and `AN`).
- Phase fields `PGT` and `PID`.
- Observation status defined by the program.
- Ternary representation of SNV substitutions.

Records with multiple alternate alleles are processed by separating the alternate alleles. Therefore, the final number of rows may depend on how multiallelic records are represented in the VCF and may not necessarily match the number of original lines in the file.

Preserving these fields makes it possible to review each variant's identity and metadata without reducing the description to a single aggregated score.

3.3.5 REF Validation Against the GRCh38 FASTA

One of the main additions in this stage is the comparison of each variant's reference allele with the corresponding base in the FASTA sequence.

The procedure retrieves the `GRCh38` reference base at the specified position and compares it with the `REF` field in the VCF. The result is recorded in two columns:

- `FASTA_REF`: base retrieved from the reference sequence.
- `REF_MATCH_FASTA`: indicator of literal agreement between the retrieved base and the `REF` allele.

This check makes it possible to identify discrepancies that require review before interpreting a molecular consequence.

For SNVs, the comparison is performed between a single reference base and the VCF base. For INDELs, the implementation compares the complete `REF` field with the FASTA base at the initial position. Therefore, the implemented indicator does not constitute a complete validation of reference alleles of variable length. For INDELs, rigorous verification must compare the complete reference sequence spanned by `REF`, while also considering record normalization.

A literal match does not, by itself, validate all aspects of a variant, such as the exact reference version, normalization, allele representation, or call quality.

3.3.6 Genomic Region Classification

Using the `GENCODE v50` annotation, the program evaluates whether each position overlaps the CDS, UTR, or exon intervals recorded for *BUB1B*.

The classification prioritizes categories in the order shown in Table 3.13.

Table 3.13: Priority order in the classification of genomic regions.

Priority	Generated category	Operational meaning
1	`CDS`	The position overlaps an annotated coding interval.
2	`UTR`	The position overlaps an annotated UTR region.
3	`EXON_NON_CDS`	The position overlaps an exon but not CDS or UTR according to the loaded intervals.
4	`INTRON_OR_GENE_SPAN`	No overlap with the preceding intervals was detected within the examined region.

The last category is deliberately broad. It does not definitively distinguish between an intron in a specific transcript and a position that simply falls within the gene's overall interval. The classification is based on the set of loaded intervals and does not represent an exhaustive annotation of all possible consequences for each transcript.

The program also records the transcripts that overlap the position. This information preserves the annotation context and facilitates subsequent review when a variant may have different consequences depending on the transcript considered.

3.3.7 Detection of Proximity to Exon-Intron Boundaries

The script includes a procedure for calculating the distance between each variant position and the start and end boundaries of annotated exons.

A configurable window of 10 bases is established. If the calculated minimum distance is equal to or less than this threshold, the variant is marked as being close to an exon boundary.

The generated fields are:

- `SPLICE_DISTANCE`: minimum distance to the nearest exon boundary.
- `NEAR_SPLICE`: proximity indicator according to the configured threshold.
- `SPLICE_BOUNDARY`: type of nearest boundary, exon start or end.
- `SPLICE_TRANSCRIPT`: transcript associated with the selected boundary.

This operation is a geometric filter for locating variants that warrant specific inspection. Proximity to an exon-intron boundary does not demonstrate that a variant alters RNA splicing.

The implemented calculation considers the boundaries of all loaded exons and records the nearest one. By itself, it does not evaluate the functional orientation of the splice site, splice-site strength, the consequence for RNA, or the experimental effect on transcript processing.

3.3.8 Coding Sequence Reconstruction

For SNVs that overlap CDS, the program attempts to reconstruct the coding sequence of each candidate transcript.

The procedure uses the CDS intervals from `GENCODE v50` and retrieves the corresponding sequences from the `GRCh38` FASTA. The segments are concatenated in genomic order and, for transcripts on the negative strand, the reverse complement operation is applied to each segment and the order of their positions is reversed.

Once the sequence has been reconstructed, the program locates the genomic position within the coding sequence and calculates:

- Base index within the CDS.
- Base position within the codon.
- Reference codon.
- Alternate codon.
- Reference amino acid.
- Alternate amino acid.
- Amino acid change represented using one-letter codes.

This procedure is limited to single-nucleotide variants with a single-base `REF` and a single-base `ALT`, for which a corresponding position is found within the reconstructed CDS sequence.

If a variant does not meet these conditions, the program leaves the coding consequence fields empty rather than assigning an uncalculated change.

3.3.9 Classification of Calculated Molecular Consequences

The code uses a codon table to translate reference and alternate codons into amino acids. Based on the comparison between them, it assigns one of the labels described in Table 3.14.

Table 3.14: Implemented classification of molecular consequences.

Label	Implemented criterion
`SYNONYMOUS`	The reference and alternate amino acids are identical.
`STOP_GAINED`	The alternate codon translates to a stop codon.
`STOP_LOST`	The reference codon is a stop codon and the alternate codon is not.
`MISSENSE`	The amino acids differ and neither of the two preceding conditions is met.

These labels describe the result of the calculation performed by the script. They do not constitute a clinical classification of the variant.

The documented implementation selects the first valid CDS annotation it finds for a variant. Therefore, the result does not guarantee that all possible consequences across the different *BUB1B* transcripts have been calculated.

Furthermore, the code does not incorporate a complete validation of each transcript's reading frame into this calculation, nor does it explicitly check all integrity conditions of the reconstructed sequence. The consequences should be considered results calculated by the program and reviewed against an appropriate transcript annotation before being used as definitive molecular nomenclature.

3.3.10 Preservation of the TriNova.OS Ternary Representation

The program preserves the ternary representation used by the `TRINOVA.OS` engine:

- Adenine (A) $\rightarrow +1$.
- Thymine (T) / Uracil (U) $\rightarrow -1$.
- Cytosine (C) and Guanine (G) $\rightarrow 0$.

For SNVs, the ternary transition between the reference and alternate alleles is recorded. INDELs and other records that do not meet the SNV definition are assigned the label `NA` for this representation.

Shannon entropy metrics in base 2 are also calculated from the ternary reference and alternate values, and from the alternate alleles of variants marked as observed.

These metrics belong to the computational framework of `TRINOVA.OS`. They are not equivalent to direct measures of pathogenicity, protein impact, or clinical causality.

3.3.11 Genotype and Phase Audit

The `audit_bub1b_track1()` block organizes the information for each variant to facilitate review before preparing the Track 1 output.

The audit gathers:

- Position and alleles.
- Variant type.
- Genotype `GT`.
- Phase fields `PGT` and `PID`.
- Depth and genotype quality.
- Recorded allele frequency.
- Annotated region.
- Calculated consequence.
- Agreement between `REF` and FASTA.

The script pays particular attention to positions 40.209.701 and 40.220.612, identified in the previous stages as heterozygous variants annotated in CDS.

For these sites, it attempts to assess whether informative phase fields are available in `PGT` and whether a shared `PID` identifier exists. If both conditions are met, it compares the phase representation of both records and generates a potential relationship label,

CIS_EVIDENCE or TRANS_EVIDENCE. If the fields are not informative, it reports that phase remains undetermined with the available information.

This logic must be interpreted precisely: the presence of PGT and PID may provide phase information, but their meaning depends on the VCF convention, the phase block, and how those fields were generated. The label produced by the program is an operational classification of the available information, not independent proof of biological phase.

In particular, two heterozygous variants are not sufficient on their own to establish that a pair is in *trans*. The audit is designed to prevent the candidate *compound heterozygous* label from being derived solely from the presence of two 0/1 genotypes.

3.3.12 Integration of Clinical Information

The program reloads clinico.json and extracts the clinical features, HPO terms, and their identifiers.

The information is preserved in a separate text file to keep the case context available during variant review.

In the documented code, this information does not contribute to the calculation of coding consequences or to the classification of variants by region. It is also not used to assign pathogenicity, causality, or clinical scores.

Therefore, the clinical integration at this stage is documentary: it allows the case data to be preserved alongside the genomic audit without asserting that a causal relationship has been established between the clinical features and the *BUB1B* records.

3.3.13 Planned Output Files

The script is configured to generate seven output files (Table 3.15).

Table 3.15: Planned output files for the Functional Zoom analysis.

File	Planned content
bub1b_funcional_todas.tsv	Table of all retrieved variants with annotation and audit fields.
bub1b_funcional_snv.tsv	Subset classified as SNVs.
bub1b_funcional_cds.tsv	Variants classified in CDS.
bub1b_funcional_splice.tsv	Variants within the configured exon-proximity window.
bub1b_funcional_summary.txt	Summary of counts, metrics, and calculated consequences.
bub1b_funcional_clinical.txt	Clinical features extracted from clinico.json.
bub1b_track1_audit.tsv	Audit table of genotypes, phase, region, and reference concordance.

The generation of these files is specified in the code. Their actual existence, final counts, and column contents must be confirmed using the output of a completed execution.

3.3.14 Methodological Scope

This stage adds checks and annotations that were not equivalently present in the initial inspection. These include comparison of REF against FASTA, regional classification based

on annotated intervals, codon calculation for certain coding SNVs, and review of available phase fields.

The scope, however, remains computational. The program does not perform an ACMG assessment, assign pathogenicity, or determine clinical causality. It also does not perform expression assays, experimental functional analyses, or independent validation of effects on RNA or protein.

The contribution of this stage is to produce a more detailed and auditable representation of the retrieved variants, explicitly documenting the available data and the checks performed by the code.

3.3.15 Documentation Status

This section documents the methodological design of `ZOOM FUNCIONAL BUB1B`, including its input resources, annotation procedures, planned calculations, phase audit, and output files.

The empirical results of this stage remain pending incorporation from the actual execution log and the generated files. Once available, the observed counts, matches or discrepancies with FASTA, calculated coding consequences, variants near exon boundaries, and retrieved phase information can be documented, consistently distinguishing program outputs from any interpretation that exceeds their scope.

3.4 Results of the Functional Audit of BUB1B

3.4.1 Execution Objective

The third analysis stage, executed using the program `trinova_os_bub1b_auditoria_completa.py`, aimed to expand the computational characterization of the variants previously identified in *BUB1B*, incorporating annotation by genomic region, preliminary assessment of consequences in the coding sequence, proximity to exon boundaries, comparison with the `GRCh38` reference, and review of the available information for determining the phase of two CDS variants.

This stage expands the variant search and characterization developed in the previous phases. Its purpose is to produce organized and auditable computational evidence for the Track 1 analysis workflow.

3.4.2 Verification of Input Files

The program confirmed the presence of the five required files:

- Variant VCF and its index.
- Gene annotation `GENCODE v50`.
- `GRCh38` reference sequence.
- Clinical information file `clinico.json`.

The analyzed sample was `WGS_PROBAND01`. During VCF traversal, the program processed 5.012.204 records and identified 14 variants within the genomic interval assigned to *BUB1B*.

3.4.3 Gene Annotation Context

The `GENCODE v50` annotation for *BUB1B* retrieved the elements detailed in Table 3.16.

Table 3.16: Gene annotation context retrieved for *BUB1B*.

Element	Count
Gene identifier	`ENSG00000156970.15`
Chromosome	`chr15`
Genomic start	40.160.984
Genomic end	40.221.137
Transcripts	31
Exonic records	511
CDS blocks	366
UTR regions	166

The exon, CDS, and UTR counts correspond to annotation records, not to numbers of unique exons or regions in the gene. The same genomic segment may be represented in multiple transcripts.

3.4.4 Summary of Detected Variants

The execution retrieved 14 variants in the *BUB1B* region. The reported distribution is summarized in Table 3.17.

Table 3.17: Summary of the distribution of variants detected in *BUB1B*.

Category	Count
Total variants	14
SNVs	9
INDELs/other	5
Variants classified in CDS	2
Variants classified as exon/CDS	2
Variants near splice boundaries (< 10 bp)	0

The 14 variants were retained in the output files, with genotype, depth, quality, and regional annotation fields. The `INDEL/otros` category is the classification used by this program and does not imply that all events have an identical molecular structure.

3.4.5 Genotype Distribution and Observed Quality

The output recorded heterozygous (`0/1`) and alternative homozygous (`1/1`) genotypes. Both CDS variants appear as heterozygous (Table 3.18).

Table 3.18: Genotypes and supporting metrics for the audited CDS variants.

Position	REF > ALT	Genotype (`GT`)	DP	GQ
40.209.701	T > G	`0/1`	46	99
40.220.612	T > G	`0/1`	28	99

DP corresponds to the reported read depth, and GQ to the genotype quality indicated in the VCF. These values describe the evidence recorded in the file and do not replace independent validation of each variant.

3.4.6 Variants Annotated in CDS Regions

The program classified two variants as belonging to coding regions (Table 3.19).

Table 3.19: Calculated consequences for variants in CDS regions.

Position	Change	Genotype	Reported consequence
40.209.701	T > G	0/1	STOP_GAINED (TTA > TGA, L > *)
40.220.612	T > G	0/1	MISSENSE (AAT > AAG, N > K)

In the main table, the first variant was annotated as a codon change introducing a stop codon, while the second was annotated as an amino acid change. These labels describe the consequences calculated by the program from the reconstructed CDS sequence.

Annotation status: both consequences should be considered preliminary. The output contains an internal discrepancy: the main summary displays the calculated codons and amino acids, but in the CDS audit block, the CODON and AA fields appear as -, even though the STOP_GAINED and MISSENSE labels are retained. Therefore, the codon and amino acid changes should not be presented as confirmed until this discrepancy is resolved and the annotation is verified against the corresponding transcript.

The review must also consider that the implemented CDS reconstruction does not explicitly incorporate the phase (*frame*) of each CDS record from the GTF. Consequently, the calculated protein consequences require additional technical verification before they can be used as definitive annotations.

3.4.7 REF Audit Against the GRCh38 Reference

The program reported four variants for which its literal REF check did not match the base retrieved from FASTA (Table 3.20).

Table 3.20: Variants with discrepancies in the literal comparison of REF against GRCh38.

Position	VCF REF	ALT	Type
40.180.642	CT	C	INDEL/other
40.181.093	AT	A	INDEL/other
40.216.568	TA	T	INDEL/other
40.216.570	TA	T	INDEL/other

This result requires precise technical interpretation. The check used by the program retrieves a single FASTA base and compares it with the complete REF field from the VCF. For an insertion/deletion variant, the REF field may contain several bases. Therefore, comparing CT with C, for example, does not constitute complete validation of the reference sequence.

Consequently, the four discrepancies are recorded as failures of the implemented literal check, but they do not, by themselves, establish that four variants are incompatible with

GRCh38. To resolve them, the complete FASTA sequence spanning the length of REF must be compared with the VCF REF, respecting the coordinates and variant representation.

For the two CDS SNV variants, the output did report a match between REF and FASTA.

3.4.8 Assessment of Proximity to Splice Boundaries

The program did not identify any variants within the ten-base threshold used to flag proximity to exon boundaries: the summary reported zero variants near a *splice* site.

This result should be expressed as a conclusion limited to the criterion applied by the program. It does not demonstrate that all intronic variants lack effects on RNA processing, nor is it equivalent to an exhaustive functional prediction of *splicing*. The tool measures proximity to annotated boundaries; it does not run a specialized model of splicing alterations.

3.4.9 Phase Assessment Between the Two CDS Variants

The analysis examined whether the variants at positions 40.209.701 and 40.220.612 could be established as occurring on different homologous chromosomes based on the PGT and PID fields in the VCF.

The results were:

- Both variants have genotype 0/1.
- Neither has an informative PGT value in the output.
- No shared PID was identified between them.
- The program classified the phase as indeterminate.

Therefore, the presence of two heterozygous variants in the same gene is not sufficient to demonstrate that they are in *trans*. Based on the information available in this execution, it cannot be established that they form a *compound heterozygous* variant pair.

This conclusion reflects a limitation in the available phase evidence, not a demonstrated absence of biological phase information. Additional reliable phase information would be required to resolve it, if such data are available.

3.4.10 TriNova.OS Entropy Metrics

The execution reported the following metrics:

- **TriNova REF/ALT Shannon:** 1,495538030
- **Observed Shannon:** 1,224394445

These values correspond to the metrics calculated by the TRINOVA.OS implementation using the base representations defined in the program. They should be interpreted as computational descriptors of the variant set, not as clinical measures of pathogenicity, disease probability, or functional effect.

Their purpose in this chapter is to document the method's quantitative output and allow it to be traced and compared with other executions performed using the same definitions.

3.4.11 Retrieved Clinical Context

The program successfully loaded `clinico.json` and retrieved eight clinical phenotype entries corresponding to the set of HPO terms used in the previous stages.

This retrieval confirms that clinical data were available to contextualize the analysis. However, the execution does not, by itself, demonstrate a genotype-phenotype association or establish that the *BUB1B* variants explain the recorded phenotypes.

3.4.12 Generated Files

The execution confirmed the generation of seven output files (Table 3.21).

Table 3.21: Output files from the complete functional audit of *BUB1B*.

File	Purpose and content
`bub1b_funcional_todas.tsv`	Annotated variant set.
`bub1b_funcional_snv.tsv`	Subset of SNV variants.
`bub1b_funcional_cds.tsv`	Variants classified in CDS.
`bub1b_funcional_splice.tsv`	Output intended for reviewing proximity to splice boundaries.
`bub1b_funcional_summary.txt`	Summary of results.
`bub1b_funcional_clinical.txt`	Retrieved clinical information.
`bub1b_track1_audit.tsv`	Audit of variants, genotypes, phase, and reported consequences.

These files constitute the reproducible artifacts of the execution and allow the results to be reviewed outside the console.

3.4.13 Summary of Results and Evidence Status

The third stage expanded the computational characterization of *BUB1B* by retrieving 14 variants, classifying them by type and region, calculating `TRINOVA.OS` metrics, and generating a specific audit of the two variants annotated in CDS. The execution identified two heterozygous CDS candidates and reported the labels `STOP_GAINED` and `MISSENSE` for them, respectively.

However, the execution log leaves three technical issues unresolved: complete validation of `REF` for the four flagged INDELs, the discrepancy between the codon/amino acid fields in the main table and those in the audit, and insufficient information to establish the phase between the two CDS variants.

These issues delimit the scope of the conclusions without invalidating the search performed. The results constitute a documented computational prioritization and annotation output; the stated uncertainties should accompany any subsequent use of the affected annotations.

3.4.14 Stage Conclusion

The functional audit of *BUB1B* produced an inventory of variants with regional annotation, genotypes, metrics, and verifiable output files. The execution identified two CDS variants for specific review, but their phase remained indeterminate, and their coding consequences require additional technical verification due to the inconsistency detected in the output.

In the context of Track 1, this stage documents the computational identification and characterization of genomic candidates. It does not assign pathogenicity, causality, or an ACMG classification, and it is not intended to replace specialized clinical interpretation. The main result is a structured and auditable set of variants and observations that can be incorporated into the project's genomic analysis with the explicitly documented technical limitations.

3.5 Integrative Summary and Conclusion of the Genomic Analysis

3.5.1 Integration of the Analysis Stages

Chapter 3 documents the genomic analysis process developed using `TRINOVA.OS`, from the architectural exploration of chromosome 15 to the detailed inspection of the variants identified in *BUB1B*.

The first stage processed 170.651 variants on chromosome 15, integrated the `GENCODE v50` annotation, and retrieved eight HPO phenotype terms from the clinical file. The engine evaluated 2.098 genes and generated 698.505 candidate pairs for the initial prioritization level. Subsequently, *BUB1B* underwent a targeted inspection independent of its position in the global ranking.

The second stage increased the resolution of the analysis by examining the genomic interval of *BUB1B*. A total of 14 variants were retrieved and organized according to their type, genotype, and relationship to the gene annotations. This phase made it possible to build a localized inventory for further computational characterization.

The third stage incorporated comparison with the `GRCh38` reference, regional classification, preliminary annotation of CDS consequences, calculation of ternary metrics, and review of the available fields for determining genetic phase. The C4 layer was integrated as an additional layer for preserving and representing molecular information, including codon changes, with the aim of retaining the detail of the results throughout the analysis process.

3.5.2 Consolidated Computational Findings

The process produced a set of candidate variants organized by gene, coordinates, alleles, and priority level. In the case of *BUB1B*, two heterozygous variants located in CDS regions were identified as highlighted candidates:

- **`chr15:40209701 T>G:`** genotype `0/1`, depth DP = 46, and genotype quality GQ = 99.
- **`chr15:40220612 T>G:`** genotype `0/1`, depth DP = 28, and genotype quality GQ = 99.

The program's functional annotation assigned the consequences `STOP_GAINED` and `MISSENSE` to these variants, respectively. The C4 layer makes it possible to preserve the codon information associated with the annotation without losing molecular detail during result integration.

However, the phase information available in the VCF did not make it possible to establish whether both variants are located on different homologous chromosomes. For

this reason, classifying the pair as biallelic compound heterozygosity must be understood as a pipeline prioritization hypothesis, not as a confirmed genetic phase.

Likewise, four variants of the INDEL/other type were flagged by the literal REF-FASTA check. Since the check used compares a single reference base with the complete `REF` field, the concordance of these records must be verified by a sequence comparison that accounts for the full length of `REF`.

These qualifications are retained as part of the analysis traceability. They do not eliminate the results obtained, but delimit which attributes are directly supported by the execution and which require additional verification.

3.5.3 Consolidation of the Final Candidate Table

Based on the search, prioritization, and result organization process, Table 3.22 was consolidated with the candidates for the Track 1 submission. The table retains the proband identifier, coordinates and alleles of each variant, the EPCR value, and the candidate type.

Note: The EPCR values and the *primary* and *secondary* categories are retained as provided by the pipeline. The table represents the project's final computational prioritization; the EPCR values are not interpreted here as calibrated clinical probabilities. The notes describe the proposed rationale for each candidate and do not constitute validation of causality.

3.5.4 Scope of the Final Table

The table consolidates five candidate findings for submission: one primary candidate associated with *BUB1B* and four secondary candidates corresponding to *GOLGA8IP*, *NIPA1*, *SNRPN*, and *ARHGAP11A-DT*.

Including a candidate in this table indicates that it was selected by the analysis process for reporting. It does not imply that all variants underwent the same level of auditing or that their genotypes, phase, functional annotations, and biological relationships were independently confirmed.

In particular, the *BUB1B* pair is retained as the primary candidate because it was prioritized by the pipeline and both variants were retrieved in the targeted analysis. The phase between them remains indeterminate based on the available PGT/PID fields. Therefore, the expression "candidate compound heterozygous pair" is more accurate than claiming that a biallelic configuration in *trans* has been demonstrated.

The four secondary candidates are included as part of the consolidated output. Their inclusion does not replace individual auditing of their coordinates, alleles, annotations, and supporting evidence, which is addressed in the following chapter.

3.5.5 Chapter Conclusion

The work presented in this chapter made it possible to progress from a chromosome-wide genomic exploration to a targeted inspection of *BUB1B* and the consolidation of a final table of five candidates for Track 1.

The main contribution of `TRINOVA.OS` is to organize the identified variants, prioritize candidates, and preserve the data required to trace their coordinates, genotypes, and annotations. The incorporation of the C4 layer expands the preservation of molecular information, including codon changes, within the analysis output.

Table 3.22: Consolidated genomic candidates for the Track 1 submission (structured vertical format).

Field / Attribute	**Finding Details**
1. Primary Candidate (*Primary*)	
Proband	PROBAND01
Variant 1	*chr15*: 40209701 (**REF:** T, **ALT:** G)
Variant 2	*chr15*: 40220612 (**REF:** T, **ALT:** G)
EPCR / Type	0,98 – **Primary**
Notes	*BUB1B*: candidate pair of heterozygous variants, stop-gained and missense; MVA. Phase not confirmed.
2. Secondary Candidates (*Secondary*)	
Proband	PROBAND01
Variant 1	*chr15*: 22587629 (**REF:** T, **ALT:** C)
Variant 2	*chr15*: 22608113 (**REF:** C, **ALT:** T)
EPCR / Type	0,85 – **Secondary**
Notes	*GOLGA8IP*: compound heterozygous pair on chromosome 15.
Proband	PROBAND01
Variant 1	*chr15*: 22814504 (**REF:** A, **ALT:** G)
Variant 2	*chr15*: 22826077 (**REF:** G, **ALT:** GA)
EPCR / Type	0,80 – **Secondary**
Notes	*NIPA1*: secondary candidate in a region relevant to neurological and developmental phenotypes.
Proband	PROBAND01
Variant 1	*chr15*: 24958486 (**REF:** GTTTTTTT, **ALT:** G)
Variant 2	*chr15*: 24974365 (**REF:** T, **ALT:** C)
EPCR / Type	0,75 – **Secondary**
Notes	*SNRPN*: secondary candidate in the 15q11-q13 region, associated with epigenetic regulation.
Proband	PROBAND01
Variant 1	*chr15*: 32573465 (**REF:** T, **ALT:** C)
Variant 2	*chr15*: 32574480 (**REF:** G, **ALT:** C)
EPCR / Type	0,65 – **Secondary**
Notes	*ARHGAP11A-DT*: secondary candidate proposed for review of its regulatory and transcriptional context.

The results are presented within their computational scope. Prioritization is not equivalent to demonstrating pathogenicity or causality, and the genetic phase of the primary *BUB1B* pair was not established using the data available in this execution. These limitations are documented to maintain the accuracy of the report and facilitate the final review of the submitted records.

With the candidate table consolidated, the search, prioritization, and genomic characterization stage is considered complete. The following chapter will focus on the final audit of the reported variants and on verifying their documentary consistency.

Transition to Chapter 4

Chapter 4. Final Variant Audit and Conclusions

Chapter 4 takes the consolidated table in Chapter 3 as its starting point. Its objective will be to review the correspondence between each reported candidate and its supporting records, check coordinates and alleles, verify the consistency of the available annotations, and document any limitations affecting the interpretation of the output.

This audit will make it possible to conclude the report with an explicit relationship between the submitted results, the computational evidence supporting them, and the checks that could be completed. The final product will be a traceable conclusion about the outcome of the variant search and prioritization process, without attributing claims to the analysis that are not supported by the data.

Chapter 4

Final Variant Audit and Conclusions

4.1 Audit Execution Results and Documentary Closure

4.1.1 Objective and Execution Context

As the concluding phase of the project and to support report traceability, the complete audit script `trinova_os_bub1b_auditoria_completa_er.py` was executed. This execution processed the project's input resources, scanned the VCF file for sample `WGS_PROBAND01` (5.012.204 records examined), and focused the audit on the genomic interval of *BUB1B* on chromosome 15 (40.160.984 – 40.221.137).

The purpose of this section is to document the actual results produced by the execution log, comparing them with the methodological design to formally close the genomic analysis of Track 1.

4.1.2 Resource Verification and Ingestion

The execution successfully confirmed the availability and integrity of the five critical system components through the following statuses:

- **VCF and its index (`.tbi`):** Verified and accessible.
- **GENCODE v50 annotation:** Loaded correctly, identifying the *BUB1B* gene (`ENSG00000156970.15`) with 31 transcripts, 511 exons, 366 CDS blocks, and 166 UTR regions.
- **Reference FASTA (GRCh38):** Contig loaded successfully for `chr15` and `15` (101.991.189 bases indexed in memory).
- **Clinical file (`clinico.json`):** Loaded correctly, retrieving 8 phenotypic features associated with HPO terms.

4.1.3 Empirical Inventory of Variants in BUB1B

The specific scan of the *BUB1B* interval retrieved a total of 14 variant records in the sample. The empirical distribution classified by structural type and region is detailed in Table 4.1.

Table 4.1: Empirical inventory of variants detected in *BUB1B* during the audit.

Parameter or category	**Recorded empirical value**
Total records in *BUB1B*	14
Variants classified as SNVs	9
Variants classified as INDEL / others	5
Records in CDS (coding) regions	2
Records in introns or control intergenic regions	10
Variants near splicing boundaries (< 10 pb)	0
Literal REF vs. FASTA discrepancies	4

4.1.4 Detailed Audit of Variants in the CDS Region

Table 4.2: Empirical parameters and calculated consequences for *BUB1B* CDS variants.

Position	**REF / ALT**	**Genotype (GT)**	**DP**	**GQ**	**REF vs. FASTA**	**Calculated consequenc**
40.209.701	T / G	0/1	46	99	YES	STOP_GAINED
40.220.612	T / G	0/1	28	99	YES	MISSENSE

Both variants showed positive literal concordance (YES) with the GRCh38 reference and high genotype quality values (GQ = 99). The calculated consequences correspond to the introduction of a premature stop codon (STOP_GAINED, codon TTA > TGA, leucine-to-stop-codon change) and a missense amino acid substitution (MISSENSE, codon AAT > AAG, asparagine-to-lysine change, N > K).

4.1.5 Assessment of REF Concordance Against GRCh38

The audit engine recorded exactly four discrepancies in the literal comparison between the VCF REF allele and the base extracted from the FASTA sequence (Table 4.3).

Table 4.3: Records with discrepancies in the literal comparison against the FASTA reference.

Position	**VCF REF**	**ALT**	**Structural type**
40.180.642	CT	C	INDEL_OTHER
40.181.093	AT	A	INDEL_OTHER
40.216.568	TA	T	INDEL_OTHER
40.216.570	TA	T	INDEL_OTHER

As anticipated in the methodological guidelines, these four differences correspond to insertions/deletions (*indels*) in which the REF field contains multiple bases, whereas the simple verification routine compared the segment against the initial base length. They do not represent structural mutations incompatible with the reference genome, but rather technical limitations of the script's literal scanning procedure that do not affect the main SNVs.

4.1.6 Phase Audit and Compound Heterozygosity Assessment

A critical finding of the empirical execution concerned the assessment of the cis/trans relationship between the two heterozygous CDS variants (40.209.701 and 40.220.612). The results of the phase-audit block were conclusive:

- Both positions have a conventional heterozygous genotype of type `0/1`.
- The phase field `PGT` returned no informative values at either position (reported as `.`).
- No shared phase-block identifier (`PID`) was identified between the two variants (the PID record was blank, `.`, for both, apart from independent blocks at other unrelated positions).
- **Resulting phase status: INDETERMINATE** based on the available `PGT/PID` fields.

Consequently, the system issued the mandatory validation directive: *"Compound heterozygosity must not be called solely because both variants have GT 0/1."* This empirically confirms that, although the pair is a computationally high-priority candidate, its biological confirmation as biallelic compound heterozygosity in *trans* remains unresolved pending long-read sequencing or familial segregation data.

4.1.7 TriNova.OS Quantitative Metrics

The execution generated the global Shannon entropy metrics corresponding to the software's mathematical model:

- **TriNova REF/ALT Shannon entropy (global interval):** 1,495538030
- **Observed Shannon entropy (ALT alleles present in the sample):** 1,224394445

The observed difference reflects the statistical divergence between the complete repertoire of theoretical substitutions accessible within the interval and those actually instantiated in the genotype of the analyzed proband.

4.1.8 Integration of Audit Output Files

The execution process successfully completed the export of the seven tabular and text artifacts to the results directory:

1. `bub1b_funcional_todas.tsv`
2. `bub1b_funcional_snv.tsv`
3. `bub1b_funcional_cds.tsv`
4. `bub1b_funcional_splice.tsv`
5. `bub1b_funcional_summary.txt`
6. `bub1b_funcional_clinical.txt`
7. `bub1b_track1_audit.tsv`

These files constitute the auditable documentary evidence that the pipeline operated transparently, preserving each intermediate parameter without altering the project's primary inputs.

4.1.9 General Conclusions of the Document

The genomic analysis developed throughout this report has made it possible to establish a reproducible, computationally rigorous, and methodologically auditable workflow using the `TRINOVA.OS` architecture.

As a final summary:

1. Chromosome 15 was explored, processing more than 170.000 variants and integrating HPO phenotypic profiles with `GENCODE v50` annotations.
2. The *BUB1B* gene was isolated, identifying 14 variants in sample `WGS_PROBAND01`.
3. Two key coding SNV variants were characterized (`chr15:40209701 T>G` and `chr15:40220612 T>G`), with estimated molecular consequences of premature termination (*stop-gained*) and amino acid substitution (*missense*).
4. Empirical auditing established that the genomic phase between the two variants is **indeterminate** due to the absence of phase markers (`PGT/PID`) in the VCF, prudently classifying the pair as a high-priority candidate rather than a confirmed biallelic configuration.
5. The final candidate table for Track 1 was consolidated, integrating the primary finding in *BUB1B* alongside four secondary candidates analyzed with the same rigor of computational traceability.

This document formally concludes with the delivery of this audit, providing a structured, transparent analytical framework free from unfounded clinical assertions and fully suitable for integration into the specialized review workflows of Track 1.

Author's Final Words and Acknowledgements

The development of this work and the `TRINOVA.OS` architecture arose from exploring large data spaces using fundamental tools built strictly from scratch. By avoiding heavy external libraries, the focus was placed on applying an original ternary encoding directly to the genomic challenge, working strictly from first principles and bare-metal data processing. Computer science and mathematics provide structure and prioritization, but biological and clinical interpretation ultimately remains in the hands of specialists who understand the living complexity of the patient.

I would like to express my deepest and most respectful gratitude to the child and their family, who chose to share their genomic data openly with the research community. This work and this hackathon would simply not exist without their immense generosity, courage, and trust.

I am also profoundly grateful to the organizers, partners, and evaluators of the MVA Hackathon 2026 for creating spaces that foster rigorous research and technology for positive human impact. Special thanks are extended to Sage Bionetworks, the MVA Society, Hugging Face, and BEACON (The Benchmarking, Evaluation, and Assessment Consortium for Science) for their organization and partnership, as well as to AWS and Anthropic for their prize sponsorship. I also extend my gratitude to the Wilhelm Foundation for their guidance and insight throughout this initiative, and to GitHub and the GENCODE Consortium for providing the open infrastructure, datasets, and tools that make reproducible computational biology possible.

I hope this work represents a meaningful contribution to this collective scientific effort.

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